SIMATS ENGINEERING

ASSESSMENT TOOL 3 - FLOW CHART TO CODE CONVERSION

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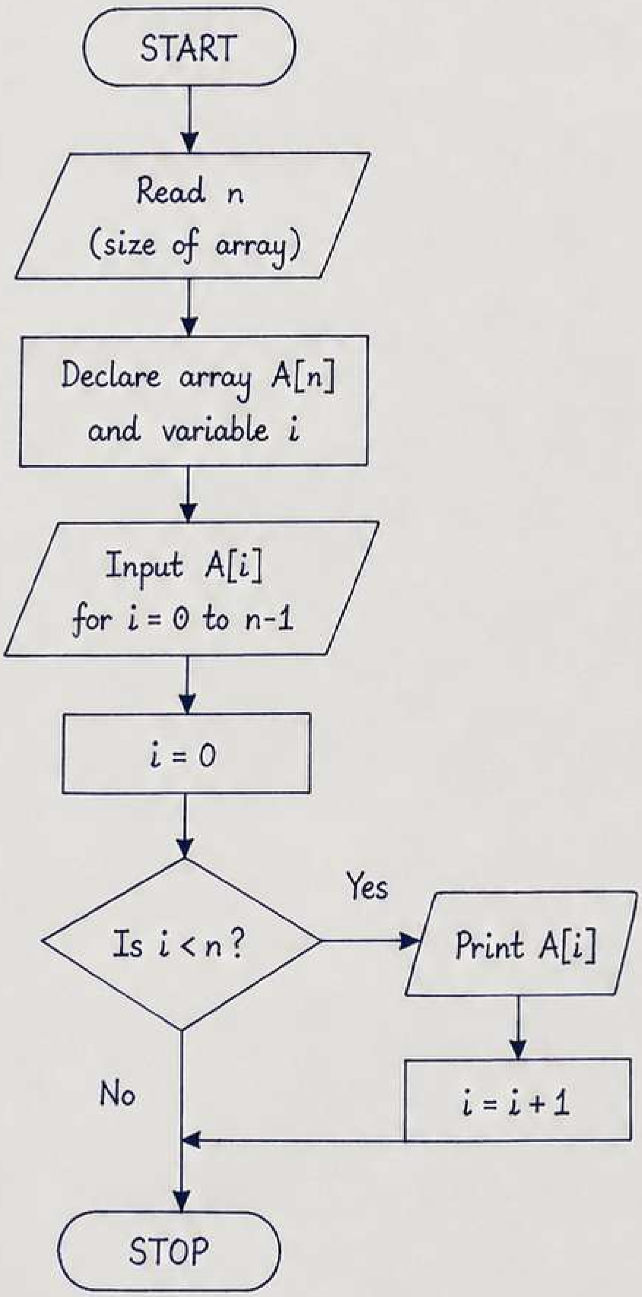
Course Code : CSA0313

1. Convert a flowchart for traversing an array and displaying all its elements into a C program.

FLOWCHART

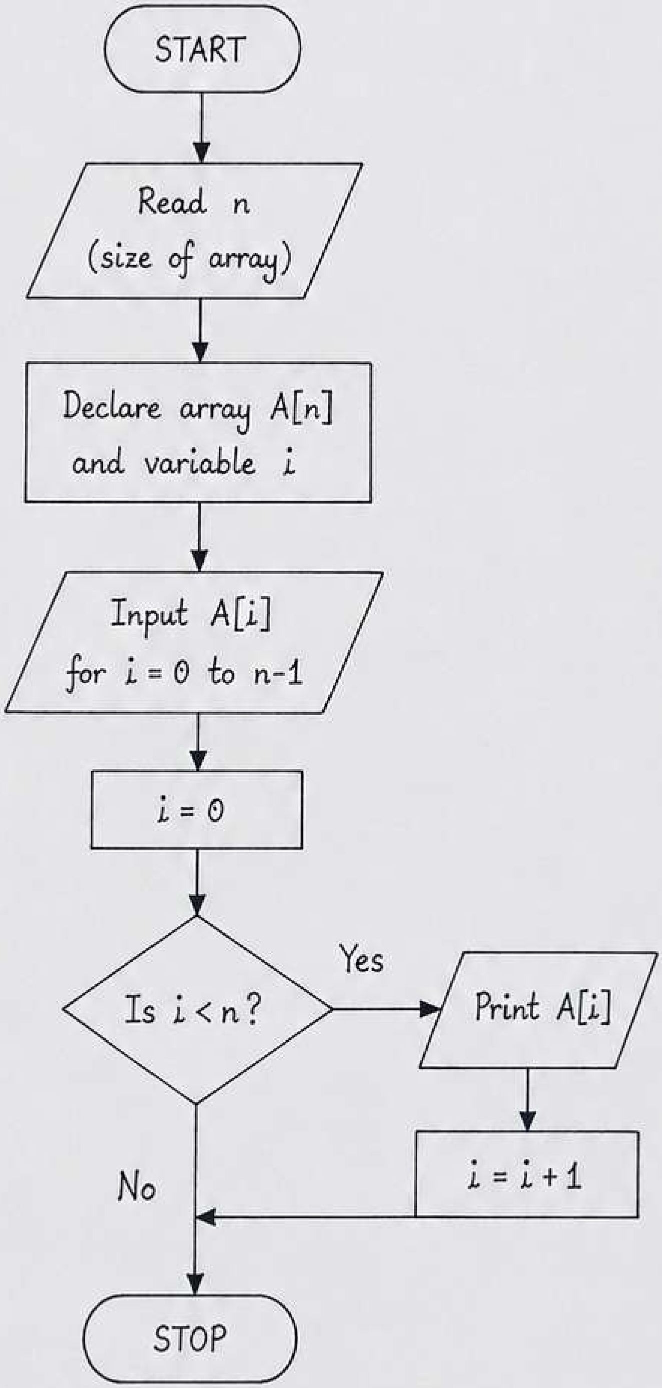


C PROGRAM

```c
#include <stdio.h>

int main()
{
    int n, i, A[100];

    printf ("Enter the size of the array: ");
    scanf ("%d", &n);

    // Input array elements
    for(i = 0; i < n; i++)
    {
        printf ("Enter element %d: ", i);
        scanf ("%d", &A[i]);
    }

    // Display all elements
    printf ("\nArray elements are:\n");
    for(i = 0; i < n; i++)
    {
        printf ("%d\t", A[i]);
    }

    printf ("\n");

    return 0;
}
```

2. <u>Convert a flowchart for traversing an array and displaying all its elements into a C program.</u>

FLOWCHART

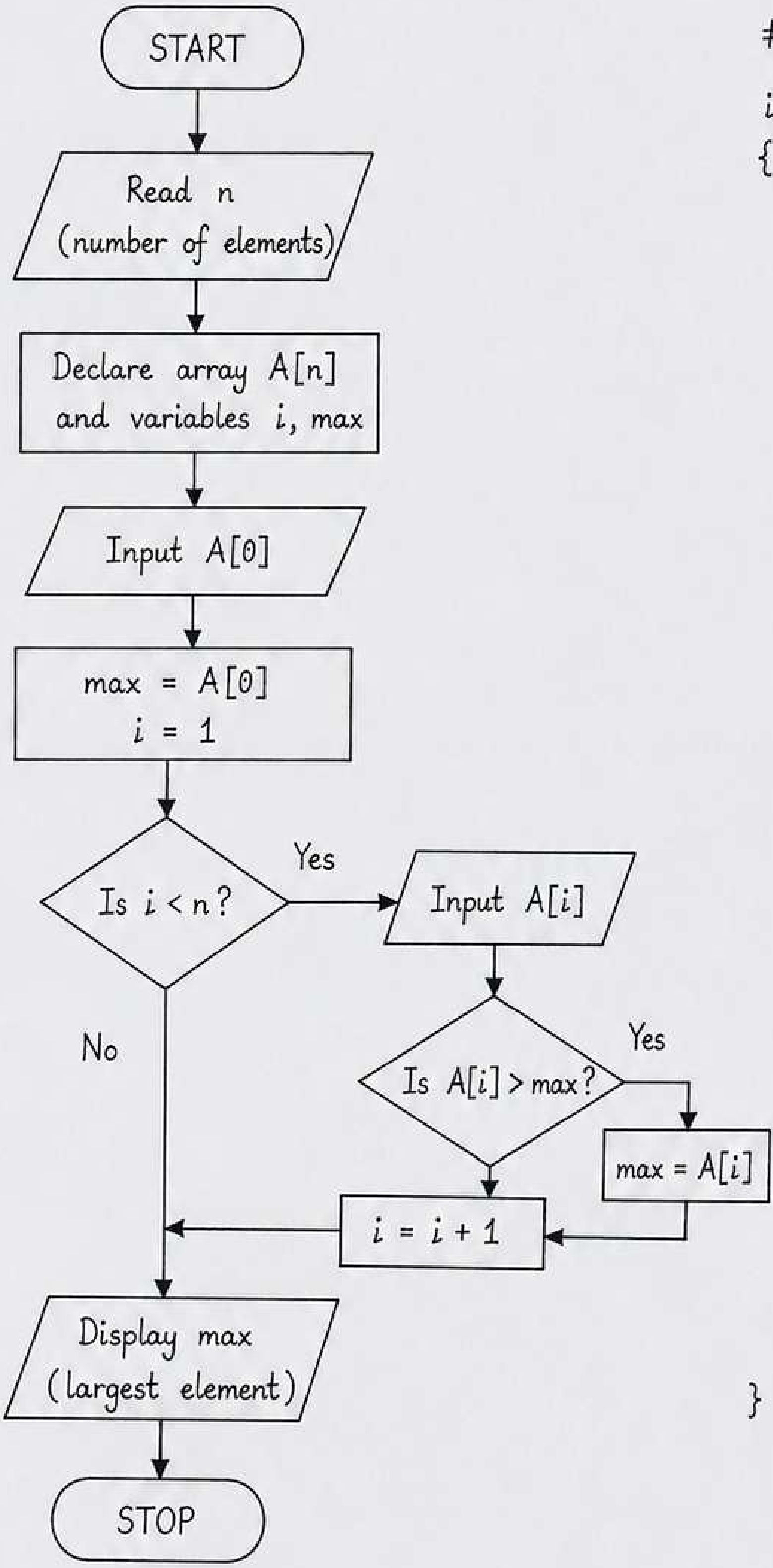


C PROGRAM

```c
#include <stdio.h>

int main()
{
    int n, i, A[100];

    printf("Enter the size of the array: ");
    scanf("%d", &n);

    // Read array elements
    for(i = 0; i < n; i++)
    {
        printf("Enter element %d: ", i);
        scanf("%d", &A[i]);
    }

    // Display all elements
    printf("\nArray elements are:\n");
    for(i = 0; i < n; i++)
    {
        printf("%d\t", A[i]);
    }

    printf("\n");

}
```

3. A flowchart accepts N elements into an array and finds the largest element. Convert the flowchart into C code.

FLOWCHART

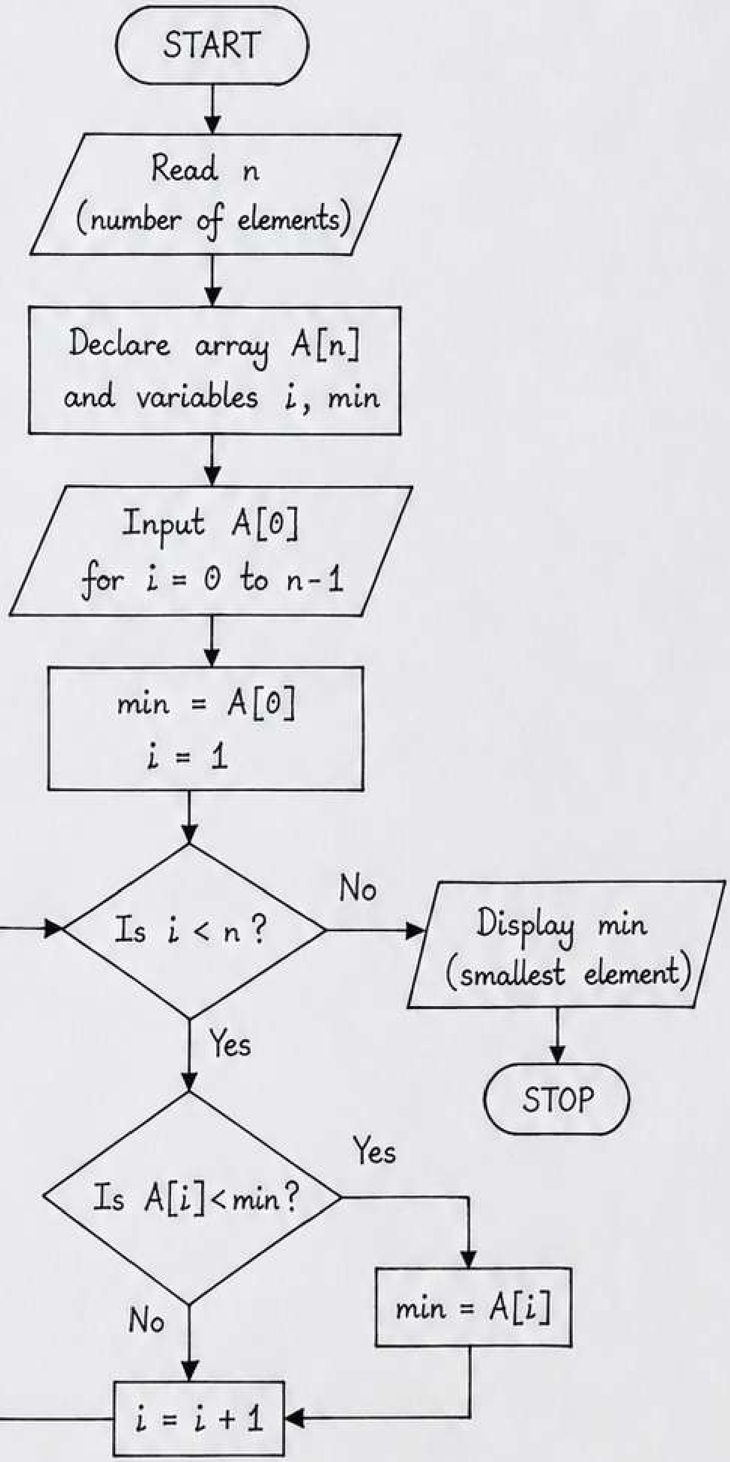


C PROGRAM

```
#include <stdio.h>

int main()
{
    int n, i, A[100], max;
    printf("Enter the number of elements: ");
    scanf("%d", &n);

    printf("Enter element 1: ");
    scanf("%d", &A[0]);
    max = A[0];
    i = 1;

    while (i < n)
    {
        printf("Enter element %d: ", i+1);
        scanf("%d", &A[i]);

        if (A[i] > max)
            max = A[i];

        i = i + 1;
    }

    printf("Largest element is: %d\n", max);

    return 0;
}
```

4. Convert a flowchart for finding the smallest element in an array into a C program.

FLOWCHART

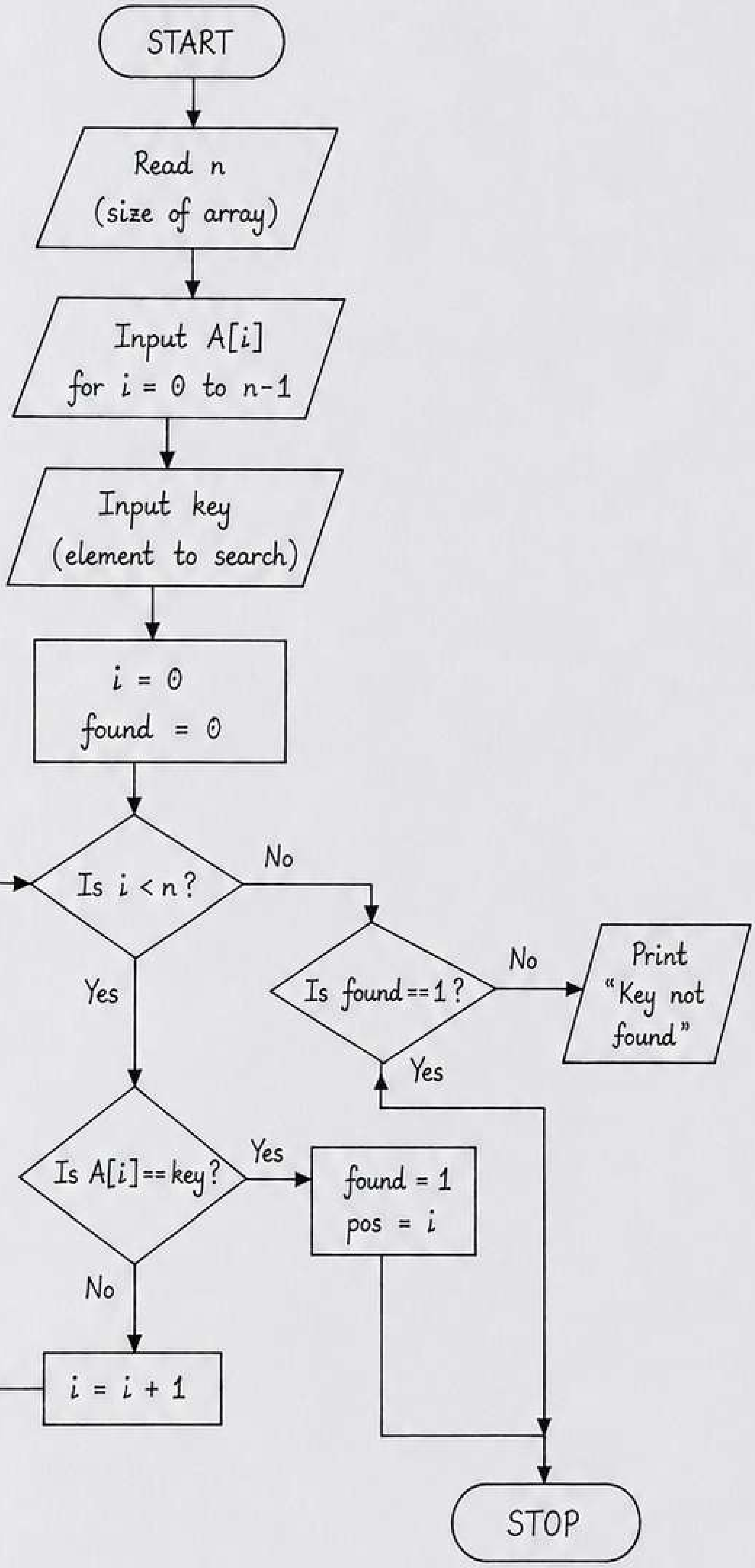


C PROGRAM

```
#include <stdio.h>

int main()
{
    int n, i, A[100], min;

    printf("Enter the number of elements:");
    scanf("%d", &n);

    printf("Enter the elements:\n");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &A[i]);
    }

    min = A[0];

    for(i = 1; i < n; i++)
    {
        if(A[i] < min)
            min = A[i];
    }

    printf("Smallest element is: %d\n", min);

    return 0;
}
```

5. A flowchart performs linear search for a given key in an array. Convert it into C code.

FLOWCHART

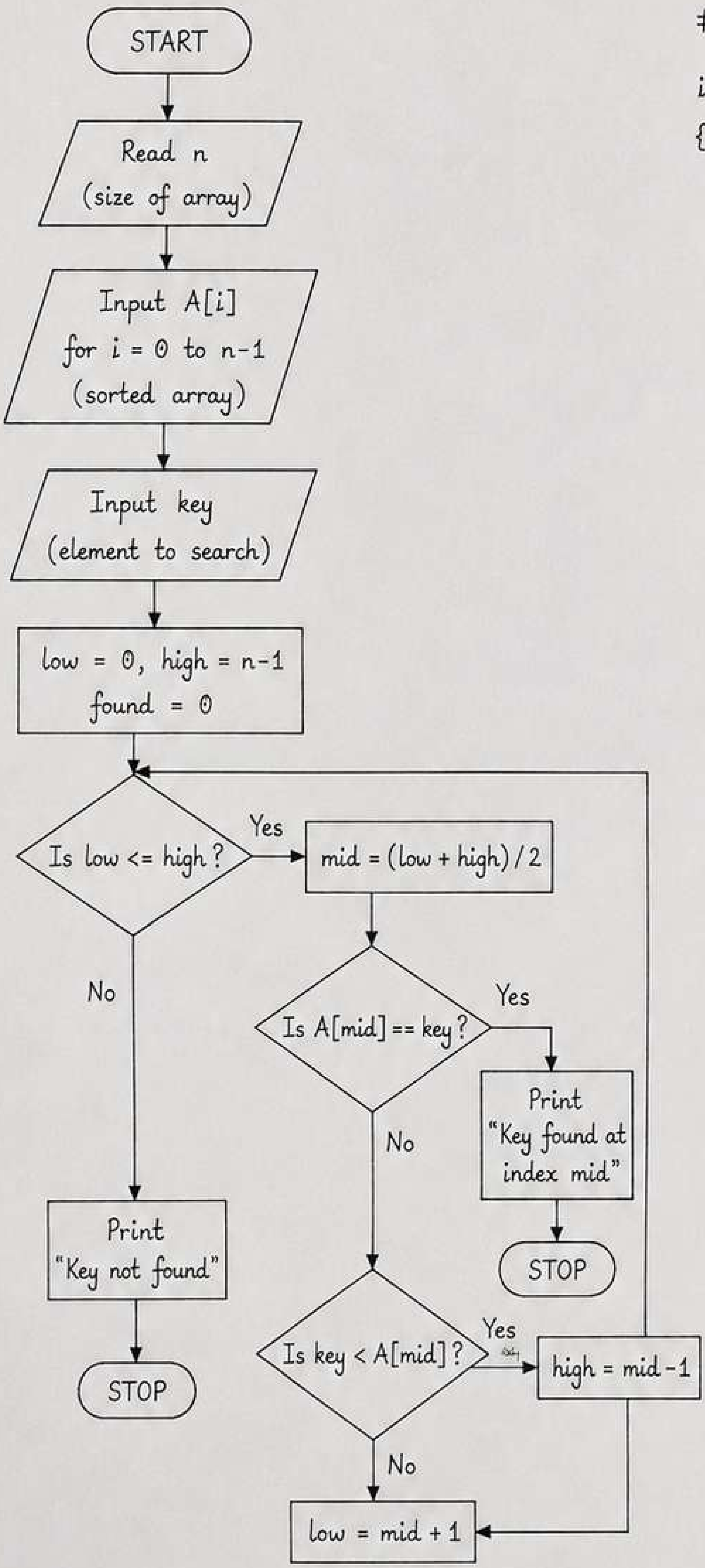


C PROGRAM

```c
#include <stdio.h>

int main()
{
    int n, i, A[100], key, found = 0, pos = -1;

    printf("Enter the size of the array: ");
    scanf("%d", &n);

    printf("Enter the elements:\n");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &A[i]);
    }

    printf("Enter the key to search: ");
    scanf("%d", &key);

    i = 0;
    while (i < n)
    {
        if(A[i] == key)
        {
            found = 1;
            pos = i;
            break;
        }
        i = i + 1;
    }

    if (found == 1)
        printf("Key found at index %d\n", pos);
    else
        printf("Key not found\n");

    return 0;
}
```

6. A flowchart performs binary search for a given key in a sorted array. Convert it into C code.

FLOWCHART

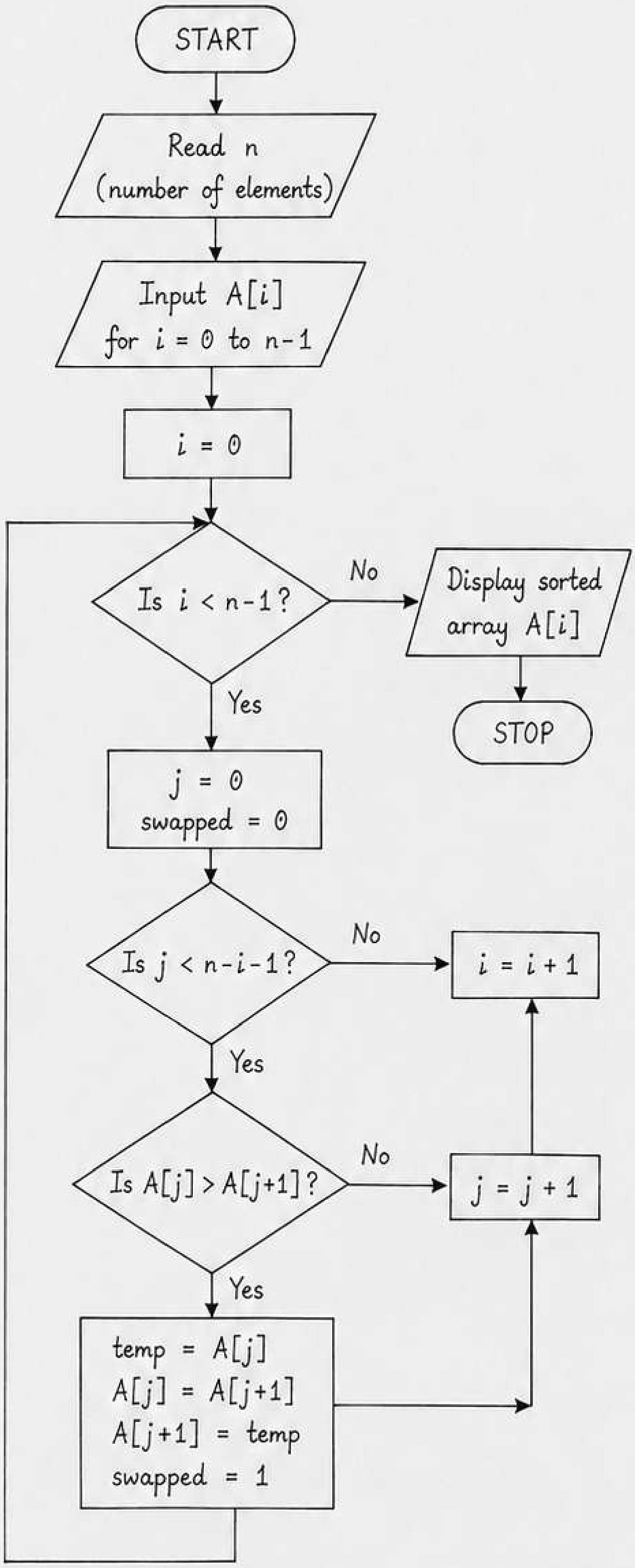


C PROGRAM

```c
#include <stdio.h>

int main()
{
    int n, i, A[100], key, low, high, mid, found = 0;
    printf("Enter the size of the array: ");
    scanf("%d", &n);

    printf("Enter the elements in sorted order:\n");
    for (i = 0; i < n; i++)
    {
        scanf("%d", &A[i]);
    }

    printf("Enter the key to search: ");
    scanf("%d", &key);

    low = 0;
    high = n - 1;

    while (low <= high)
    {
        mid = (low + high) / 2;

        if (A[mid] == key)
        {
            printf("Key found at index %d\n", mid);
            found = 1;
            break;
        }
        else if (key < A[mid])
            high = mid - 1;
        else
            low = mid + 1;
    }

    if (found == 0)
        printf("Key not found\n");

    return 0;
}
```

7. Convert this bubble-sort flowchart for an array of N elements into C code.

FLOWCHART

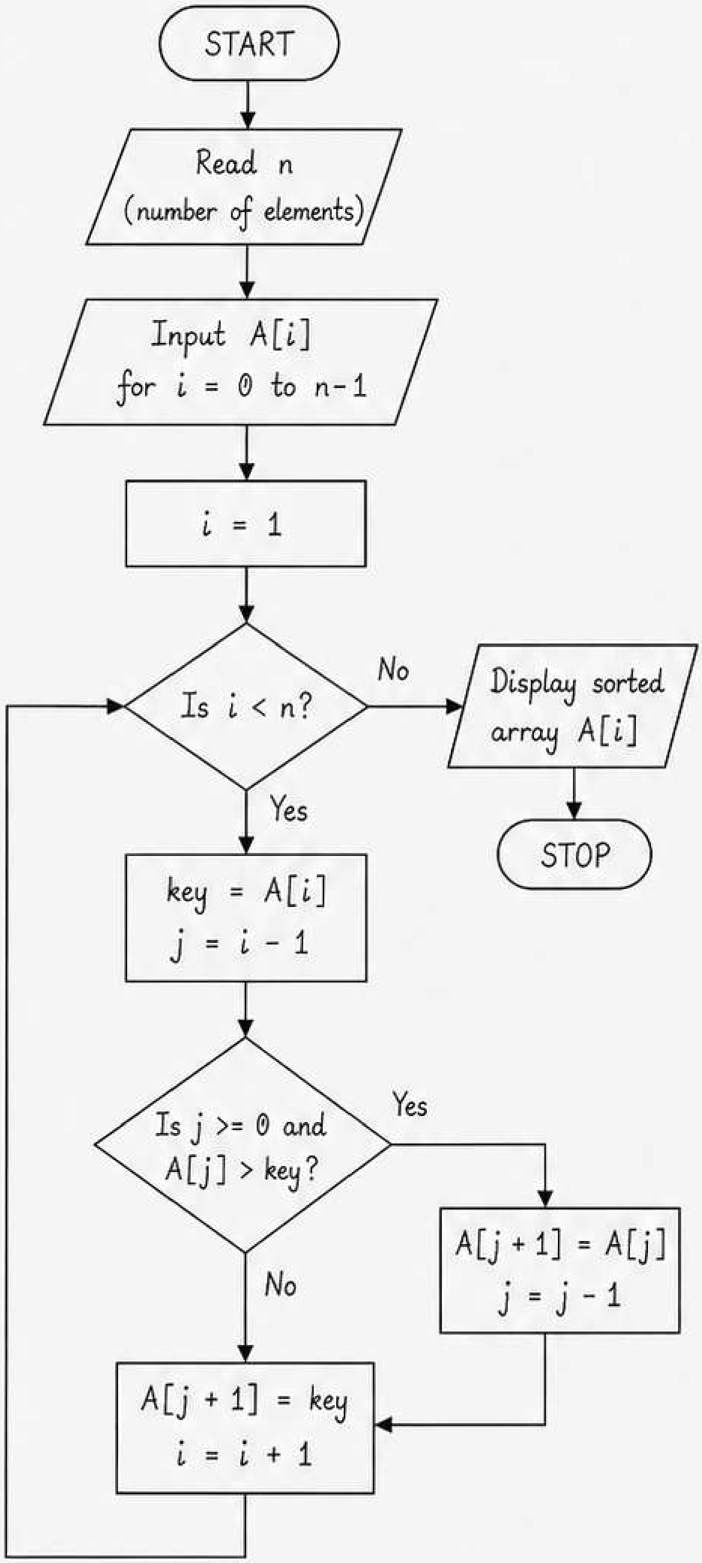


C PROGRAM

```c
#include <stdio.h>

int main()
{
    int n, i, j, A[100], temp, swapped;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    printf("Enter the elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &A[i]);

    for(i = 0; i < n - 1; i++)
    {
        swapped = 0;
        for(j = 0; j < n - i - 1; j++)
        {
            if(A[j] > A[j + 1])
            {
                temp = A[j];
                A[j] = A[j + 1];
                A[j + 1] = temp;
                swapped = 1;
            }
        }
    }

    printf("\nSorted array:\n");
    for(i = 0; i < n; i++)
        printf("%d\t", A[i]);
    printf("\n");

    return 0;
}
```

8. A flowchart performs insertion sort on an array of N elements. Convert it into C code.

FLOWCHART

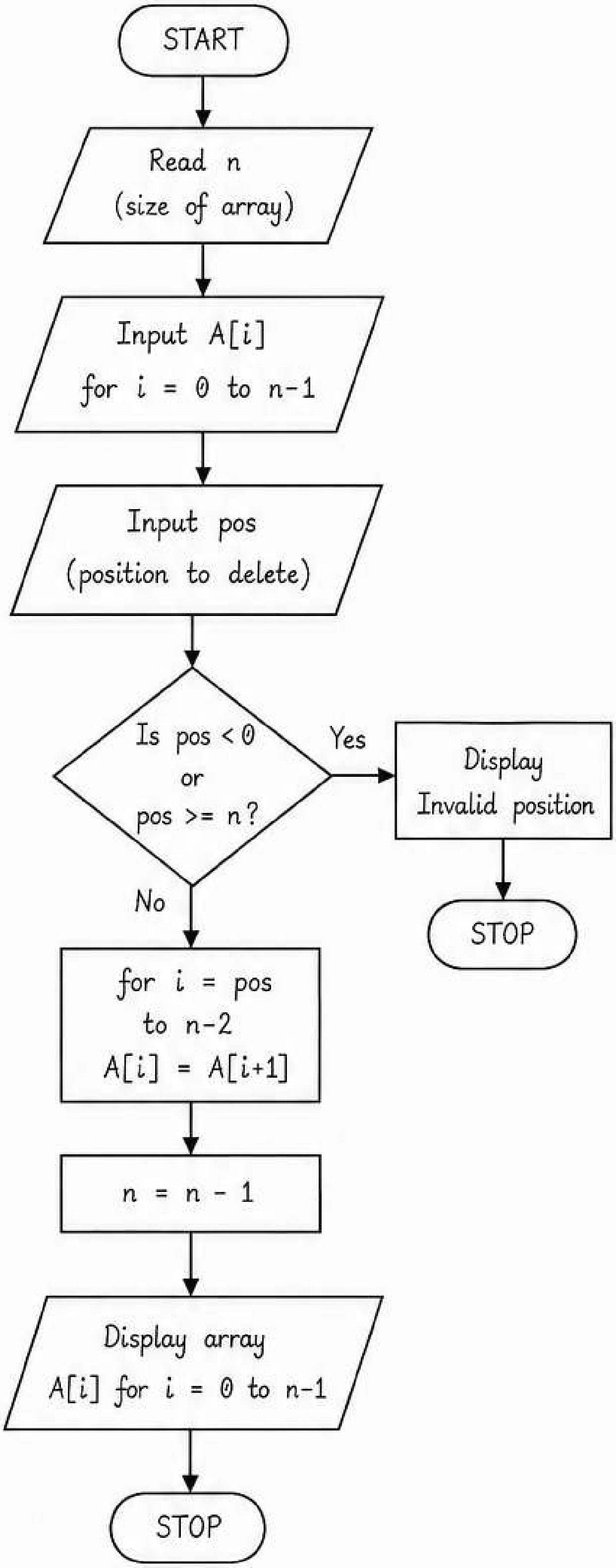


C PROGRAM

```
#include <stdio.h>

int main()
{
    int n, i, j, A[100], key;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    printf("Enter the elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &A[i]);

    for(i = 1; i < n; i++)
    {
        key = A[i];
        j = i - 1;

        while (j >= 0 && A[j] > key)
        {
            A[j + 1] = A[j];
            j = j - 1;
        }
        A[j + 1] = key;
    }

    printf("\nSorted array:\n");
    for(i = 0; i < n; i++)
        printf("%d\t", A[i]);
    printf("\n");

    return 0;
}
```

9. A flowchart performs deletion from an array at a given position. Convert it into C code.

FLOWCHART

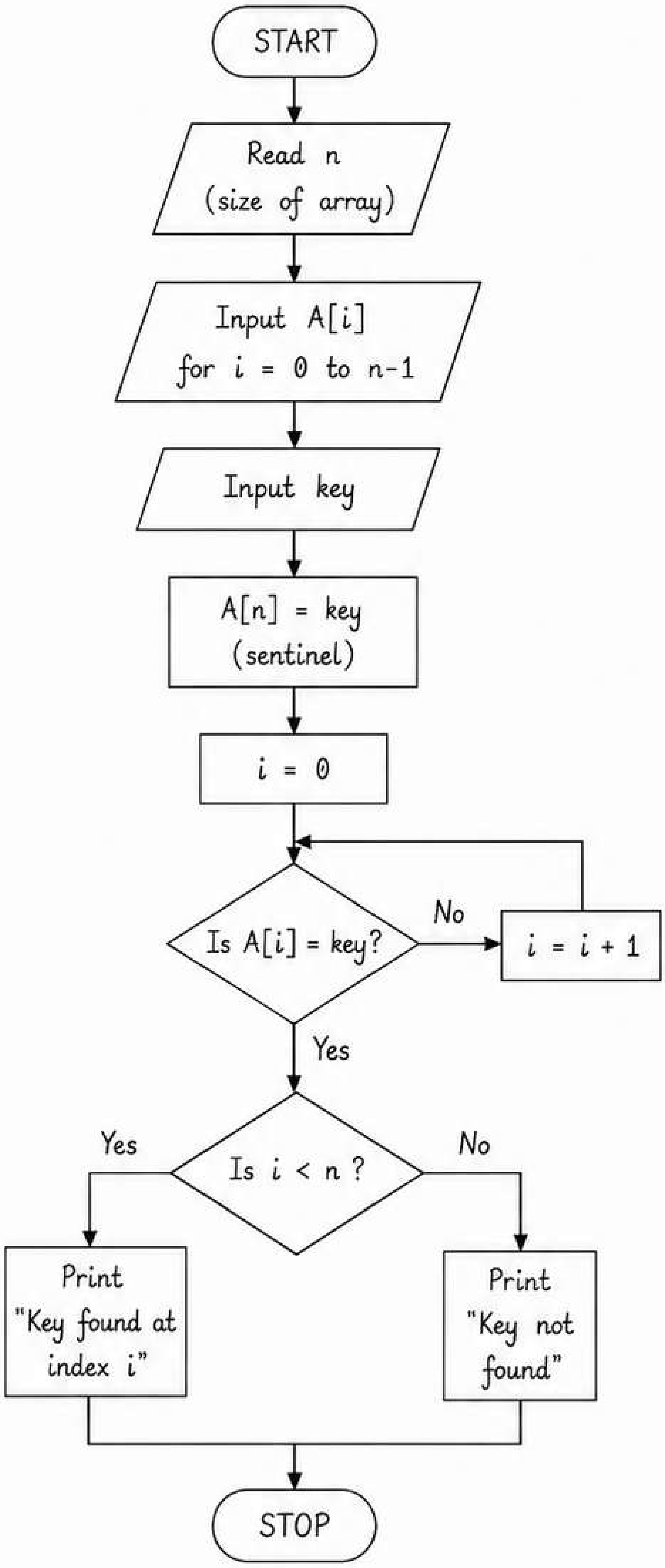


C PROGRAM

```c
#include <stdio.h>

int main()
{
    int n, i, pos, A[100];

    printf("Enter the size of the array: ");
    scanf("%d", &n);

    printf("Enter the elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &A[i]);

    printf("Enter the position to delete: ");
    scanf("%d", &pos);

    if (pos < 0 || pos >= n)
    {
        printf("Invalid position\n");
        return 0;
    }

    for (i = pos; i < n - 1; i++)
    {
        A[i] = A[i + 1];
    }

    n = n - 1;

    printf("\nArray after deletion:\n");
    for (i = 0; i < n; i++)
        printf("%d\t", A[i]);
    printf("\n");

    return 0;
}
```

10. A flowchart performs linear search for a given key in an array using a sentinel. Convert it into C code.

FLOWCHART



C PROGRAM

```
#include <stdio.h>

int main()
{
    int n, i, A[100], key;

    printf(“Enter the size of the array: ”);
    scanf(“%d”, &n);

    printf(“Enter the elements:\n”);
    for (i = 0; i < n; i++)
        scanf(“%d”, &A[i]);

    printf(“Enter the key to search: ”);
    scanf(“%d”, &key);

    A[n] = key;        // sentinel
    i = 0;

    while (A[i] != key)
        i = i + 1;

    if (i < n)
        printf(“Key found at index %d\n”, i);
    else
        printf(“Key not found\n”);

    return 0;
}
```